Output Tokens
(shifted right)

[S]	the	dog	and	cat	play		

	t	0	1	2	3	4	5	6	7
[S]	0	1	0	0	0	0	0	0	0
a	1	0	0	0	0	0	0	0	0
and	2	0	0	0	1	0	0	0	0
apples	3	0	0	0	0	0	0	0	0
brings	4	0	0	0	0	0	0	0	0
by	5	0	0	0	0	0	0	0	0
cat	6	0	0	0	0	1	0	0	0
day	7	0	0	0	0	0	0	0	0
dog	8	0	0	1	0	0	0	0	0
eat	9	0	0	0	0	0	0	0	0
fill	10	0	0	0	0	0	0	0	0
friend	11	0	0	0	0	0	0	0	0
happy	12	0	0	0	0	0	0	0	0
house	13	0	0	0	0	0	0	0	0
laughter	14	0	0	0	0	0	0	0	0
light	15	0	0	0	0	0	0	0	0
music	16	0	0	0	0	0	0	0	0
night	17	0	0	0	0	0	0	0	0
play	18	0	0	0	0	0	1	0	0
quiet	19	0	0	0	0	0	0	0	0
smile	20	0	0	0	0	0	0	0	0
the	21	0	1	0	0	0	0	0	0
to	22	0	0	0	0	0	0	0	0
water	23	0	0	0	0	0	0	0	0

Word Embedding

[S]	a	and	apples	brings	by	cat	day	dog	eat	fill	friend	happy	house	laughter	light	music	night	play	quiet	smile	the	to	water
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
3	0	2	1	2	3	2	2	3	0	2	2	0	2	2	2	0	0	0	2	2	0	3	3
0	3	0	3	0	2	2	1	1	3	3	1	0	1	0	3	2	0	1	1	1	3	3	0
0	1	1	2	3	1	1	3	0	1	0	0	3	1	3	2	1	3	1	0	3	2	3	0
1	1	3	3	3	3	2	1	3	2	0	2	2	3	2	3	1	2	3	1	2	3	2	2
3	1	0	3	0	2	3	2	2	2	3	2	1	3	2	3	2	3	1	2	1	0	1	2
1	3	2	1	1	2	1	2	1	1	1	2	2	3	3	1	3	1	2	0	3	1	2	1

3	0	3	2	2	0	0	0
0	3	1	0	2	1	0	0
0	2	0	1	1	1	0	0
1	3	3	3	2	3	0	0
3	0	2	0	3	1	0	0
1	1	1	2	1	2	0	0

Positional Encoding

d \ t	0	1	2	3	4	5	6	7
0	0.7768	0.8942	0.9504	0.9769	0.9892	0.995	1	0.9989
1	0.9784	0.8006	0.5911	0.4177	0.2894	0.1987	0.1	0.0927
2	-0.4556	0.1774	0.5828	0.7983	0.9046	0.9553	1	0.9903
3	0.4045	0.9594	0.9536	0.759	0.554	0.3894	0.3	0.1846
4	-0.9651	-0.6816	-0.0103	0.4738	0.7426	0.8776	0.9	0.9732
5	-0.8112	0.3491	0.9471	0.9616	0.7712	0.5646	0.4	0.2749

4	0.894198	4	2.977	2.989	0.995	0.998	0.999
1	3.800613	2	0.418	2.289	1.199	0.136	0.093
-0	2.177376	1	1.798	1.905	1.955	0.979	0.99
1	3.959425	4	3.759	2.554	3.389	0.269	0.185
2	-0.68164	2	0.474	3.743	1.878	0.943	0.973
0	1.349127	2	2.962	1.771	2.565	0.397	0.275

Transformer Decoder

Wq

Head						
0	-1	-1	1	1	0	1
	1	-1	-1	0	0	-1
	-1	-1	0	0	0	1
1	0	1	1	1	1	1
	0	1	0	1	0	1
	-1	1	0	-1	1	0
2	1	0	0	0	1	1
	-1	1	-1	1	0	1
	0	0	-1	-1	0	1

Wk

Head						
0	-1	-1	1	1	0	1
	1	-1	-1	0	1	-1
	-1	-1	0	1	0	1
1	0	1	-1	-1	0	1
	0	1	0	1	0	0
	0	1	1	0	-1	0
2	1	0	1	0	0	-1
	0	1	0	1	0	1
	0	0	1	-1	0	1

Wv

Head						
0	1	1	-1	0	1	1
	0	1	0	1	0	0
	0	-1	0	1	1	0
1	-1	1	1	0	0	1
	-1	0	0	1	0	-1
	0	1	1	0	0	0
2	1	1	0	0	1	1
	0	1	0	1	1	0
	1	1	1	0	-1	2

Multi-head Self-Attention

Q

-4	2.791117	1	5.124	0.951	5.716	0.512	0.358
3	-6.43292	-0	-2.2	-2.98	-4.72	-0.51	-0.36
-5	-3.34568	-4	-0.43	-3.51	0.371	-0.74	-0.82
4	10.6049	10	9.41	12.26	10.99	2.724	2.516
3	9.109166	7	7.138	6.614	7.153	0.802	0.552
-2	-1.73465	-4	-5.84	0.489	-1.31	-0.19	-0.12
6	1.561688	8	6.412	8.503	5.437	2.338	2.247
-1	6.037591	3	2.363	1.721	4.202	-1.17	-1.44
-1	-4.78767	-3	-2.6	-2.69	-2.78	-0.85	-0.9

K

-4	2.791117	1	5.124	0.951	5.716	0.512	0.358
5	-7.11456	2	-1.73	0.767	-2.85	0.428	0.614
-3	0.613741	0	3.326	-0.95	3.76	-0.47	-0.63
0	-0.98706	-1	-2.18	-0.4	-1.58	-0.72	-0.81
2	7.760038	6	4.177	4.843	4.588	0.405	0.277
-2	6.659627	0	1.742	0.451	1.276	0.172	0.11
3	1.722448	3	1.814	3.123	0.386	1.579	1.714
3	9.109166	7	7.138	6.614	7.153	0.802	0.552
-2	-0.43292	-1	1.001	1.122	1.131	1.108	1.081

V

7	3.184925	9	5.032	8.888	4.681	1.494	1.349
2	7.760038	6	4.177	4.843	4.588	0.405	0.277
2	-0.52283	4	3.815	4.007	4.068	1.076	1.065
-3	6.432918	0	2.201	2.976	4.724	0.515	0.359
-3	1.7161	-2	-2.18	-2.21	-0.17	-1.13	-1.09
1	5.977989	2	2.216	4.194	3.154	1.115	1.083
7	5.362301	9	6.83	10.79	6.636	2.474	2.34
4	7.078401	8	4.65	8.586	6.466	1.348	1.25
3	10.25208	8	10.64	6.983	7.401	1.965	1.659

Casual Mask

t	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	-99	0	0	0	0	0	0	0
2	-99	-99	0	0	0	0	0	0
3	-99	-99	-99	0	0	0	0	0
4	-99	-99	-99	-99	0	0	0	0
5	-99	-99	-99	-99	-99	0	0	0
6	-99	-99	-99	-99	-99	-99	0	0
7	-99	-99	-99	-99	-99	-99	-99	0

Head
0

Q

-3.6175	2.79112	0.94194	5.12431	0.95114	5.71572	0.51231	0.36
3.06516	-6.4329	-0.1706	-2.2007	-2.9759	-4.7236	-0.5148	-0.4
-4.5663	-3.3457	-3.5944	-0.433	-3.5074	0.37097	-0.736	-0.8

K^T

-3.6174744	5.1	-3.2
2.79111701	-7.1	0.61
0.94194011	1.82	0.36
5.12430844	-1.7	3.33
0.95113517	0.77	-1
5.71572381	-2.8	3.76
0.5123102	0.43	-0.5
0.35820721	0.61	-0.6

K^T * Q / sqrt(dk) + casual mask

24.9165	-18.664	4.09187	-16.392	-4.3462	-26.524	-1.2423	-0.3
-119.04	29.736	0.94505	17.1436	12.5136	28.7449	2.67946	1.76
-98.695	-104.93	-0.4123	0.38576	-3.3355	-1.7756	-0.4147	-0.4
-121.53	-90.753	-102.95	16.523	-0.9543	22.332	0.61559	-0.2
-97.116	-98.473	-96.58	-96.922	1.13585	0.84363	0.45862	0.49
-125.89	-86.483	-103.41	-79.414	-98.586	27.4289	0.93859	-0
-98.082	-98.861	-97.795	-97.911	-98.508	-98.576	0.2228	0.24
-96.995	-99.483	-97.554	-98.563	-98.579	-99.628	-98.808	0.24

V

7.43449	3.18493	8.89559	5.03161	8.88774	4.68056	1.49436	1.35
2.3829	7.76004	5.54471	4.17668	4.84333	4.58809	0.40499	0.28
2.46096	-0.5228	4.35211	3.8151	4.00717	4.06833	1.07586	1.07

t	0	1	2	3	4	5	6	7
	1	9.55638E-22	0.948720719	1.77381E-15	4.76269E-08	7.82466E-25	0.012682799	0.054936
	0	1	0.040784073	0.650346332	0.999986959	0.787488541	0.640332489	0.438335
	0	3.26925E-59	0.010495208	3.43021E-08	1.30865E-07	4.37909E-14	0.029015096	0.052926
	0	4.70048E-53	3.10068E-47	0.349653634	1.41568E-06	0.001291736	0.081297885	0.064708
	0	2.08703E-56	1.80332E-44	1.88445E-50	1.1447E-05	6.01029E-13	0.069487691	0.122472
	0	3.36408E-51	1.93879E-47	7.56437E-43	5.62269E-49	0.211219723	0.112294257	0.07514
	0	1.41567E-56	5.35035E-45	7.00748E-51	6.07956E-49	3.99547E-56	0.054889784	0.095387
	0	7.60239E-57	6.80661E-45	3.65262E-51	5.66628E-49	1.39471E-56	5.38227E-45	0.096095

7	3.184925059	7.276508955	3.830625023	3.184993903	3.503218517	4.026080617	4.313295
2	7.760038476	2.635380723	6.50710253	7.75999947	7.085431147	6.373659151	5.09933
2	-0.522825108	2.35912073	0.993947247	-0.522766332	0.452521192	0.927219718	1.384666

Head
1

Q

4.15096	10.6049	10.0643	9.41032	12.2617	10.9856	2.7242	2.52
2.57172	9.10917	7.49186	7.13824	6.6145	7.15273	0.80248	0.55
-2.168	-1.7346	-4.3232	-5.8444	0.48872	-1.3082	-0.1885	-0.1

K^T

0.21836339	2.38	-1.5
-0.9870614	7.76	6.66
-0.9980591	5.54	0.18
-2.1780521	4.18	1.74
-0.398039	4.84	0.45
-1.581443	4.59	1.28
-0.7150188	0.4	0.17
-0.8073117	0.28	0.11

K^T * Q / sqrt(dk) + casual mask

5.95407	15.3834	15.3501	16.1092	10.2192	12.3675	1.612	1.18
-98.179	28.0983	11.2076	4.147	24.5261	20.7558	1.31823	0.59
-93.39	-76.135	17.7241	16.8071	14.161	16.4282	0.97912	0.31
-100.2	-92.115	-97.938	-0.4989	1.02277	2.11785	-1.6801	-2
-93.328	-76.417	-81.49	-82.725	15.8056	17.1356	1.56881	0.94
-97.575	-85.832	-91.53	-92.99	-92.314	7.95265	-0.5005	-0.9
-100.33	-101.42	-101.83	-101.8	-102.47	-101.99	-0.9557	-0.9
-100.66	-102.59	-102.77	-102.61	-103.63	-103.06	-100.15	-1.1

V

-3.0652	6.43292	0.17061	2.20066	2.97589	4.72364	0.51483	0.36
-2.5611	1.7161	-1.944	-2.1795	-2.2065	-0.1702	-1.126	-1.1
0.52277	5.97799	2.17388	2.21598	4.19397	3.15401	1.11502	1.08

t	0	1	2	3	4	5	6	7
	1	3.00608E-06	0.085062291	0.332273364	6.11667E-07	0.000218731	0.288262423	0.317884
	0	0.999996994	0.00135104	2.12025E-06	0.999804697	0.961344867	0.214884963	0.17595
	0	5.39713E-46	0.913586669	0.667724495	3.15087E-05	0.012689533	0.153084443	0.132651
	0	6.19742E-53	5.35995E-51	2.03557E-08	6.20238E-11	7.73643E-09	0.010715906	0.0139
	0	4.0689E-46	7.45736E-44	3.96576E-44	0.000163183	0.025744216	0.276078266	0.249003
	0	3.3182E-50	3.25426E-48	1.38095E-48	1.8069E-51	2.64583E-06	0.034861362	0.038909
	0	5.63144E-57	1.09039E-52	2.06637E-52	7.0425E-56	4.71884E-54	0.022112637	0.0389
	0	1.74099E-57	4.29259E-53	9.13479E-53	2.21059E-56	1.62608E-54	1.83532E-45	0.032802

-3	6.432889311	-0.096168375	-0.904535573	6.432150608	6.262371674	1.546102559	1.167333
-3	1.716087051	-1.99153042	-2.149023568	1.715341874	1.567731535	-1.330437801	-1.43592
1	5.977972858	2.038573137	1.625269336	5.977574935	5.882587955	3.08428161	2.783112

Head
2

Q

6.00046	1.56169	7.88722	6.41223	8.50299	5.43723	2.3377	2.25
-0.7494	6.03759	2.95869	2.36306	1.72064	4.20239	-1.1744	-1.4
-0.76	-4.7877	-2.5892	-2.5957	-2.6874	-2.7801	-0.8509	-0.9

K^T

3.13230528	2.57	-1.7
1.72244754	9.11	-0.4
2.58602073	7.49	-1.4
1.81361188	7.14	1
3.12268235	6.61	1.12
0.38569818	7.15	1.13
1.57937979	0.8	1.11
1.71433441	0.55	1.08

K^T * Q / sqrt(dk) + casual mask

10.4721	16.4085	21.1549	17.6094	20.525	18.7551	3.3049	2.8
-96.784	34.5025	24.0509	19.4533	18.1767	28.2031	-3.6389	-5.1
-92.658	-66.618	26.7017	21.9286	22.3468	28.5803	-0.8901	-2.1
-96.245	-75.249	-80.044	14.953	14.4417	21.4059	-2.8838	-4.1
-91.536	-76.228	-75.158	-80.096	20.1602	24.0505	-0.8213	-2
-101.25	-76.844	-86.715	-89.508	-91.755	16.7504	-4.8846	-6
-94.362	-97.84	-92.093	-93.718	-92.168	-93.873	1.04348	0.81
-93.774	-98.516	-91.866	-93.519	-91.712	-94.013	-97.591	1.2

V

6.97886	5.3623	9.47835	6.82991	10.7923	6.6359	2.47354	2.34
4.41777	7.0784	7.53437	4.65046	8.5859	6.46567	1.34753	1.25
2.64231	10.2521	8.02893	10.6422	6.98299	7.40071	1.96514	1.66

t	0	1	2	3	4	5	6	7
	1	1.38636E-08	0.003629528	0.012119234	0.12539814	3.18574E-05	0.878184373	0.737114
	0	0.999999986	0.065699912	0.076600733	0.011979329	0.404024439	0.000847102	0.000274
	0	1.21333E-44	0.93067056	0.910429248	0.775267952	0.589135422	0.013235431	0.005383
	0	2.16582E-48	4.07114E-47	0.000850785	0.000285977	0.00045128	0.001802396	0.000752
	0	8.13123E-49	5.38919E-45	4.47044E-45	0.087068602	0.006352711	0.014177273	0.005958
	0	4.39262E-49	5.15659E-50	3.65624E-49	2.16604E-50	4.29126E-06	0.000243745	0.000109
	0	3.34488E-58	2.38229E-52	5.42778E-51	1.43369E-50	3.88663E-54	0.09150968	0.100692
	0	1.70067E-58	2.99E-52	6.61927E-51	2.26112E-50	3.37743E-54	1.33306E-44	0.149719

7	5.362301228	9.198852343	9.130510867	9.229260216	7.822425098	6.652003973	5.866228
4	7.078401077	7.493096939	7.459213968	7.228818327	7.35541951	4.240326203	3.677157
3	10.252079	8.155444125	8.136170682	7.289771847	8.921499778	2.735198014	2.490315

7	3.184925059	7.276508955	3.830625023	3.184993903	3.503218517	4.026080617	4.313295

2	7.760038476	2.635380723	6.50710253	7.75999947	7.085431147	6.373659151	5.09933
2	-0.522825108	2.35912073	0.993947247	-0.522766332	0.452521192	0.927219718	1.384666
-3	6.432889311	-0.096168375	-0.904535573	6.432150608	6.262371674	1.546102559	1.167333
-3	1.716087051	-1.99153042	-2.149023568	1.715341874	1.567731535	-1.330437801	-1.43592
1	5.977972858	2.038573137	1.625269336	5.977574935	5.882587955	3.08428161	2.783112
7	5.362301228	9.198852343	9.130510867	9.229260216	7.822425098	6.652003973	5.866228
4	7.078401077	7.493096939	7.459213968	7.228818327	7.35541951	4.240326203	3.677157
3	10.252079	8.155444125	8.136170682	7.289771847	8.921499778	2.735198014	2.490315

9	-17.82346505	1.460124187	1.267553668	-10.99300351	-12.79151312	0.213641507	0.810134
23	4.916701821	20.76047285	20.30898318	11.89761693	9.847196196	17.73024784	16.50316
-18	-8.348875923	-16.77741201	-16.01093605	-15.32830068	-12.98265927	-15.06937224	-13.6313
3	6.519390967	7.814835157	13.34664485	7.424653883	6.525813493	2.887799093	1.491249
-19	2.161699256	-13.26737591	-12.84976922	-4.668798602	-2.491776686	-13.48992163	-12.826
17	18.95811758	22.13220994	21.13914238	19.71244078	20.42967622	16.47383527	15.47668

Linear

0	0	1	-1	0	-1	1	0	-1
1	1	1	0	-1	-1	1	1	-1
-1	-1	-1	0	-1	1	-1	-1	1
-1	0	-1	-1	0	0	1	0	1
-1	-1	-1	0	1	1	-1	0	1
1	1	1	0	0	0	1	-1	1

Add & Norm

+

13	-16.9293	5	4.244	-8	-11.8	1.211	1.809
24	8.717315	22	20.73	14.19	11.05	17.87	16.6
-18	-6.1715	-16	-14.2	-13.4	-11	-14.1	-12.6
4	10.47882	12	17.11	9.979	9.915	3.157	1.676
-17	1.480062	-11	-12.4	-0.93	-0.61	-12.5	-11.9
18	20.30724	24	24.1	21.48	22.99	16.87	15.75

Normalize

1	-1.50771	-0	-0.14	-0.88	-1.11	-0.06	-0.01
1	0.434439	1	0.84	0.761	0.556	1.148	1.156
-1	-0.69305	-1	-1.24	-1.28	-1.05	-1.18	-1.14
0	0.567833	0	0.625	0.45	0.474	0.078	-0.02
-1	-0.11362	-1	-1.13	-0.36	-0.29	-1.06	-1.08
1	1.312114	1	1.04	1.3	1.428	1.076	1.09
1	1	1	1	1	1	1	1

Scale & Shift

gamma						beta
1	0	0	0	0	0	2
0	2	0	0	0	0	3
0	0	4	0	0	0	0
0	0	0	5	0	0	1
0	0	0	0	4	0	4
0	0	0	0	0	5	2

3	0.492288	2	1.86	1.122	0.89	1.937	1.994
5	3.868877	5	4.68	4.522	4.112	5.296	5.313
-5	-2.77221	-5	-4.95	-5.11	-4.21	-4.7	-4.57
1	3.839163	3	4.123	3.25	3.369	1.392	0.916
-1	3.545519	-0	-0.51	2.58	2.823	-0.25	-0.32
6	8.560572	7	7.202	8.498	9.138	7.379	7.45

Multi-head Cross-Attention

Wq

Head						
0	0	0	1	1	0	1
	1	0	0	0	0	0
	0	0	0	0	0	1
1	0	1	1	1	1	1
	0	1	0	1	0	1
	1	0	0	0	-1	0

Q

2	9.627521	5	6.377	6.637	8.292	4.068	3.796
3	0.492288	2	1.86	1.122	0.89	1.937	1.994
6	8.560572	7	7.202	8.498	9.138	7.379	7.45
6	17.04192	10	10.54	13.74	15.23	9.11	8.786
12	16.26861	15	16	16.27	16.62	14.07	13.68
3	-3.05323	2	2.371	-1.46	-1.93	2.191	2.316

2	0.558511	2	0.885	2.347	1.026
4	2.133213	4	2.941	4.398	2.969
1	3.036335	-1	3.485	-0.48	3.471
0	0.576289	1	0.522	1.066	0.327
-0	2.986149	-1	0.704	-0.79	0.912

Wk

0	1	1	1	0	1
	0	1	0	1	0
	1	0	0	1	1
1	0	0	1	0	1
	0	1	0	1	0
	1	0	1	0	0

K

7	8.714209	5	8.014	5.47	8.377
5	2.709502	6	3.463	5.464	3.296
2	4.12095	3	2.111	2.62	2.264
0	6.022485	-2	4.189	-1.28	4.382
5	2.709502	6	3.463	5.464	3.296
3	3.594847	2	4.369	1.864	4.496

Wv

0	1	0	0	0	1
	0	1	1	0	0
	0	1	0	1	0
1	1	1	0	0	1
	1	0	1	1	0
	0	1	1	1	0

V

2	3.54466	2	1.588	1.554	1.937
5	5.169548	4	6.426	3.916	6.44
5	2.709502	6	3.463	5.464	3.296
6	5.677873	6	4.53	5.952	4.906
3	4.171136	3	4.891	2.931	4.823
5	5.745837	5	6.948	4.982	6.767

Head 0

Q

1.8644	9.62752	4.78898	6.37663	6.63706	8.29182	4.06816	3.8
2.50967	0.49229	1.96365	1.8601	1.12239	0.89036	1.93697	1.99
5.82149	8.56057	7.35887	7.20178	8.49775	9.13751	7.37893	7.45

K^T

6.92858878	4.72	2.24
8.71420873	2.71	4.12
5.2737338	5.55	2.88
8.01447698	3.46	2.11
5.46969631	5.46	2.62
8.37699548	3.3	2.26

K^T * Q / sqrt(dk)

21.8318	50.9323	34.0321	39.8976	40.6058	47.4208	31.1019	30.3
27.1567	69.5752	44.6743	52.1263	55.366	64.8505	41.0538	39.9
23.3982	45.1162	33.1061	37.347	37.9273	43.2845	30.8591	30.3
20.7388	55.9636	35.0528	42.0006	43.3097	51.2821	31.6885	30.6
22.612	44.9068	32.4508	36.9002	37.3559	42.8174	30.1207	29.5
21.4038	58.692	36.5194	43.7955	45.3457	53.7437	33.0086	31.9

t 0	1	2	3	4	5	6	7
0	8.00726E-09	2.38771E-05	4.88655E-06	3.8881E-07	2.69381E-08	4.7616E-05	6.23E-05
1	0.999979995	0.9996083	0.999713539	0.999949276	0.999983682	0.99949109	0.999431
0	2.3855E-11	9.45838E-06	3.81355E-07	2.66976E-08	4.3055E-10	3.73512E-05	6.69E-05
0	1.22617E-06	6.62604E-05	4.00274E-05	5.80771E-06	1.28033E-06	8.56081E-05	9.02E-05
0	1.93492E-11	4.91179E-06	2.4393E-07	1.50773E-08	2.69872E-10	1.78493E-05	3.06E-05
0	1.87708E-05	0.000287192	0.000240922	4.44854E-05	1.50105E-05	0.000320485	0.000319

V

1.89078	3.54466	1.70656	1.58847	1.55417	1.93743
5.03781	5.16955	3.56717	6.426	3.91552	6.43957
4.72072	2.7095	5.55499	3.46348	5.46417	3.29595

3	3.544627864	3.54400259	3.544185658	3.544576863	3.544633769	3.54379495	3.543684
5	5.16957366	5.169971813	5.169902989	5.169611963	5.169568949	5.169974359	5.169913
3	2.709514023	2.709808923	2.709685127	2.709533441	2.709511899	2.710005786	2.710157

Head 1

Q

6.49453	17.0419	9.61977	10.5449	13.7385	15.2275	9.1101	8.79
12.1176	16.2686	15.0025	16.0042	16.2698	16.6183	14.0675	13.7
3.12147	-3.0532	2.0713	2.37139	-1.4574	-1.933	2.19128	2.32

K^T

0.1775011	4.72	3.05
6.02248457	2.71	3.59
-1.5309692	5.55	1.6
4.18851897	3.46	4.37
-1.2752054	5.46	1.86
4.38249132	3.3	4.5

K^T * Q / sqrt(dk)

39.1868	40.7123	45.5213	48.8745	43.1862	43.4514	43.1319	42.3
48.0166	78.3687	61.2165	66.6233	70.1964	74.9319	58.2308	56.8
36.0142	34.2846	41.5313	44.2043	38.6867	38.0477	39.0943	38.3
47.8101	66.0411	58.4874	63.4847	62.0805	65.1785	55.6879	54.4
36.8063	35.4898	42.476	45.278	39.6435	39.1347	40.0307	39.2
47.5946	66.1519	58.2657	63.2918	61.9384	65.1345	55.5084	54.3

t 0	1	2	3	4	5	6	7
0	4.42605E-17	1.36568E-07	1.81445E-08	1.85929E-12	2.12893E-14	2.42145E-07	4.29E-07
0	0.999990626	0.894800678	0.926712578	0.999442392	0.999886321	0.873856212	0.845611
0	7.15335E-20	2.52671E-09	1.70015E-10	2.06668E-14	9.58006E-17	4.27143E-09	7.78E-09
0	4.42783E-06	0.058405895	0.040167448	0.000298587	5.80908E-05	0.068717523	0.083704
0	2.38727E-19	6.4983E-09	4.97518E-10	5.38018E-14	2.84085E-16	1.08955E-08	1.97E-08
0	4.94628E-06	0.046793282	0.033119956	0.000259021	5.55885E-05	0.057426007	0.070684

V

6.26081	5.67787	6.09028	4.52991	5.9522	4.90632
3.39952	4.17114	2.7757	4.89112	2.93051	4.82334
5.3885	5.74584	4.73845	6.94805	4.98166	6.76662

5	5.67786442	5.574722422	5.606209011	5.677330706	5.677763744	5.554681471	5.527248
5	4.171142305	4.243705919	4.221656808	4.171519803	4.17121397	4.258064747	4.277502
6	5.745847853	5.863819265	5.827935403	5.746460849	5.745964062	5.887069656	5.918621

3	3.544627864	3.54400259	3.544185658	3.544576863	3.544633769	3.54379495	3.543684
5	5.16957366	5.169971813	5.169902989	5.169611963	5.169568949	5.169974359	5.169913
3	2.709514023	2.709808923	2.709685127	2.709533441	2.709511899	2.710005786	2.710157
5	5.67786442	5.574722422	5.606209011	5.677330706	5.677763744	5.554681471	5.527248
5	4.171142305	4.243705919	4.221656808	4.171519803	4.17121397	4.258064747	4.277502
6	5.745847853	5.863819265	5.827935403	5.746460849	5.745964062	5.887069656	5.918621

1	2.64153059	2.420712079	2.487958734	2.640403298	2.64131158	2.377617602	2.318785
13	12.05022999	12.12348665	12.10124492	12.05066521	12.05029482	12.13804489	12.15757
16	15.08656382	15.277497	15.2194952	15.08759262	15.08674698	15.31510876	15.36604
8	7.879084409	7.894539097	7.889958756	7.879214691	7.879094037	7.897956177	7.902185
5	3.612611297	3.833099434	3.765912051	3.613707006	3.612834087	3.876183135	3.935057
5	5.006256146	5.077899587	5.056157339	5.006563226	5.006335841	5.091853911	5.111028

Linear

0	0	1	1	0	-1
0	1	1	0	1	0
0	1	0	0	1	1
-1	0	0	1	0	1
1	0	0	-1	0	1
1	0	-1	0	1	0

Add & Norm

+

4	3.133819	4	4.348	3.763	3.532	4.315	4.312
18	15.91911	17	16.78	16.57	16.16	17.43	17.47
11	12.31435	10	10.27	9.976	10.87	10.61	10.8
9	11.71825	11	12.01	11.13	11.25	9.29	8.818
4	7.15813	4	3.255	6.193	6.436	3.622	3.612
11	13.56683	12	12.26	13.5	14.14	12.47	12.56

Normalize

-1	-1.60723	-1	-1.06	-1.37	-1.46	-1.02	-1.01
2	1.132161	1	1.352	1.361	1.223	1.506	1.507
0	0.359802	0	0.088	-0.05	0.1	0.191	0.23
-0	0.23208	0	0.426	0.2	0.18	-0.06	-0.15
-1	-0.74497	-1	-1.28	-0.85	-0.84	-1.16	-1.14
0	0.628159	1	0.473	0.707	0.794	0.549	0.567
1	1	1	1	1	1	1	1

Scale & Shift

gamma						beta
1	0	0	0	0	0	2
0	2	0	0	0	0	3
0	0	4	0	0	0	0
0	0	0	5	0	0	1

1	0.392773	1	0.937	0.63	0.543	0.976	0.989
6	5.264321	6	5.704	5.722	5.445	6.012	6.013
1	1.439208	0	0.35	-0.18	0.402	0.762	0.919
1	2.160402	2	3.129	2.002	1.9	0.678	0.257

0	0	0	0	4	0	4
0	0	0	0	0	5	2

-0	1.020102	-1	-1.1	0.592	0.637	-0.63	-0.58
3	5.140796	5	4.367	5.534	5.972	4.745	4.837
1	1	1	1	1	1	1	1

Feed-Forward

Linear

-1	1	-1	1	1	-1	0
1	1	0	1	0	1	0
-1	0	1	-1	1	1	-1
0	1	0	0	-1	0	-1
-1	-1	0	0	1	-1	0
0	-1	1	0	1	1	1
1	1	0	1	1	1	0
-1	0	0	0	1	0	-1
-1	1	0	1	-1	0	1
0	1	0	0	-1	1	-1
1	0	1	0	0	-1	0
1	0	0	0	1	-1	0

1	1.472049	1	2.077	2.334	1.066	-0.42	-1.05
11	12.95829	13	14.14	13.89	13.86	12.41	12.1
2	4.04693	0	-1.45	2.312	3.568	2.223	2.932
5	3.24422	6	5.806	4.13	3.808	5.642	5.591
-11	-9.77779	-12	-12.1	-11.3	-11.3	-12.4	-12.4
-1	3.335784	-1	-1.09	1.221	2.566	-0.13	0.165
11	13.97839	13	13.03	14.48	14.5	11.78	11.52
-2	-0.37267	-3	-3.04	-1.04	-0.91	-2.61	-2.57
7	7.011849	9	9.998	7.503	7.165	7.344	6.859
9	8.385015	10	10.17	9.664	9.78	10.39	10.43
-1	-3.30882	-4	-3.08	-5.09	-5.03	-3.01	-2.93
-3	-3.72792	-5	-4.53	-4.31	-4.79	-4.4	-4.43

ReLU

1	1.472049	1	2.077	2.334	1.066	0	0
11	12.95829	13	14.14	13.89	13.86	12.41	12.1
2	4.04693	0	0	2.312	3.568	2.223	2.932
5	3.24422	6	5.806	4.13	3.808	5.642	5.591
0	0	0	0	0	0	0	0
0	3.335784	0	0	1.221	2.566	0	0.165
11	13.97839	13	13.03	14.48	14.5	11.78	11.52
0	0	0	0	0	0	0	0
7	7.011849	9	9.998	7.503	7.165	7.344	6.859
9	8.385015	10	10.17	9.664	9.78	10.39	10.43
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1

Linear

0	1	0	0	0	0	1	-1	1	-1	0	-1	0
0	-1	0	0	0	0	1	-1	1	0	0	-1	-1
-1	1	0	1	1	-1	0	1	1	-1	0	1	0
1	1	1	1	-1	1	-1	-1	1	0	0	-1	-1
1	-1	0	0	-1	0	-1	-1	-1	0	-1	1	0
0	-1	0	1	0	-1	1	1	0	1	0	1	-1

20	25.56352	24	27	26.2	25.74	21.15	20.05
6	7.031951	7	7.896	7.094	6.802	5.714	5.281
14	10.02151	16	17.69	12.3	11.42	15.01	13.95
15	17.09073	16	17.98	15.91	16.54	14.84	15.12
-28	-32.4765	-34	-35.1	-33.5	-34.5	-31.5	-30.5
13	8.313553	14	13.88	12.16	10.66	14.4	14.28

ReLU

20	25.56352	24	27	26.2	25.74	21.15	20.05
6	7.031951	7	7.896	7.094	6.802	5.714	5.281
14	10.02151	16	17.69	12.3	11.42	15.01	13.95
15	17.09073	16	17.98	15.91	16.54	14.84	15.12
0	0	0	0	0	0	0	0
13	8.313553	14	13.88	12.16	10.66	14.4	14.28

Add & Norm

+

21	25.95629	25	27.93	26.83	26.29	22.13	21.04
12	12.29627	13	13.6	12.82	12.25	11.73	11.29
15	11.46072	16	18.04	12.12	11.82	15.77	14.87
15	19.25113	17	21.11	17.91	18.44	15.52	15.38
-0	1.020102	-1	-1.1	0.592	0.637	-0.63	-0.58
17	13.45435	19	18.24	17.7	16.63	19.14	19.11

Normalize

1	1.442943	1	1.192	1.404	1.401	1.027	0.975
-0	-0.19282	-0	-0.28	-0.21	-0.25	-0.28	-0.29
0	-0.29287	0	0.178	-0.29	-0.3	0.23	0.176
0	0.640012	0	0.493	0.375	0.48	0.198	0.242
-2	-1.54312	-2	-1.78	-1.62	-1.61	-1.83	-1.83
0	-0.05414	0	0.199	0.35	0.268	0.653	0.726
1	1	1	1	1	1	1	1

Scale & Shift

gamma						beta
1	0	0	0	0	0	2
0	2	0	0	0	0	3
0	0	4	0	0	0	0
0	0	0	5	0	0	1
0	0	0	0	4	0	4
0	0	0	0	0	5	2

3	3.442943	3	3.192	3.404	3.401	3.027	2.975
3	2.614362	2	2.445	2.574	2.508	2.443	2.422
1	-1.1715	1	0.712	-1.17	-1.18	0.919	0.702
2	4.200062	2	3.465	2.874	3.4	1.988	2.208
-3	-2.17248	-3	-3.14	-2.49	-2.43	-3.32	-3.32
4	1.729292	4	2.994	3.751	3.342	5.265	5.63
1	1	1	1	1	1	1	1

Linear

0	0	2	0	0	1	1	0
1	1	0	-1	-2	0	1	-3
2	-1	2	-1	-2	-2	0	0
3	1	0	0	1	0	1	-4
4	0	1	-2	0	0	0	0
5	-2	-1	0	1	-2	1	2
6	0	2	1	-2	0	0	0
7	-2	0	1	0	0	1	-1
8	1	-1	2	-1	0	0	0
9	1	-1	-1	0	-1	0	-1
10	1	0	2	1	-2	0	0
11	0	0	1	-2	1	1	2
12	2	0	2	1	2	0	0
13	0	0	2	-2	1	2	0
14	2	1	-2	-2	0	-2	1
15	2	-2	2	0	-2	1	0
16	0	1	-1	0	2	-1	0
17	1	0	0	-2	0	0	0
18	1	1	1	2	2	0	0
19	-2	0	1	-2	-2	2	1
20	1	2	0	-1	-1	0	0
21	0	1	0	1	0	0	1
22	0	-1	0	-1	0	-2	0
23	-2	2	0	1	0	0	-1

Logit (Z)

6	4.785538	6	4.745	6.409	5.927	6.834	7.156
-1	-5.05639	-1	-4.45	-0.42	-1.87	0.397	0.488
4	-1.09789	3	0.336	2.153	0.862	3.601	3.388
6	5.372297	6	5.651	6.029	6.143	6.28	6.814
1	4.957361	1	1.021	4.921	4.873	0.605	1.018
6	2.774062	6	5.908	4.225	4.296	7.391	8.103
2	-4.3429	1	-1.33	-1.77	-2.97	1.83	1.131
-2	-7.32809	-2	-3.68	-5.23	-5.64	-0.87	-0.62
0	-5.71448	-0	-1.29	-4.39	-4.87	0.434	-0.25
2	3.172558	2	2.175	3.494	3.507	1.983	2.17
14	9.64496	13	14.36	8.914	9.3	13.49	13.22
-1	-8.01481	-1	-4.36	-3.66	-5.07	0.89	0.598
3	4.397993	4	4.995	2.353	2.973	3.244	2.926
2	-9.45702	2	-2.66	-3.08	-4.91	5.074	4.931
-5	0.984538	-5	-4.51	-0.52	-0.81	-6.85	-7.71
14	5.388409	13	12.19	8.046	7.627	14.91	14.78
-9	-2.28839	-9	-7.54	-4.99	-4.52	-10.4	-10.5
-1	-4.95718	-2	-3.74	-2.34	-3.4	-0.95	-1.44
4	8.940974	5	7	5.57	6.663	3.729	3.878
6	-7.65397	5	0.666	-0.24	-2.24	9.055	9.234
10	6.644082	9	7.758	8.17	7.449	9.245	8.931
6	7.814424	6	6.91	6.448	6.908	5.432	5.63
-13	-10.273	-13	-11.9	-13	-12.6	-15	-15.9
0	1.5429	-0	0.971	0.214	0.614	-0.18	0.102

Softmax

Prediction (Y)
(Output Probabilities)

	t	0	1	2	3	4	5	6	7
[S]	0	0	0.004411	0	6E-05	0.037	0.021	2E-04	4E-04
a	1	0	2.35E-07	0	6E-09	4E-05	9E-06	4E-07	5E-07
and	2	0	1.23E-05	0	7E-07	5E-04	1E-04	1E-05	9E-06
apples	3	0	0.007932	0	1E-04	0.026	0.026	1E-04	3E-04
brings	4	0	0.005238	0	1E-06	0.008	0.007	5E-07	9E-07
by	5	0	0.00059	0	2E-04	0.004	0.004	4E-04	0.001
cat	6	0	4.79E-07	0	1E-07	1E-05	3E-06	2E-06	1E-06
day	7	0	2.42E-08	0	1E-08	3E-07	2E-07	1E-07	2E-07
dog	8	0	1.21E-07	0	1E-07	8E-07	4E-07	4E-07	2E-07
eat	9	0	0.000879	0	5E-06	0.002	0.002	2E-06	3E-06
fill	10	1	0.568792	0	0.895	0.457	0.622	0.194	0.173
friend	11	0	1.22E-08	0	7E-09	2E-06	4E-07	7E-07	6E-07
happy	12	0	0.002994	0	8E-05	6E-04	0.001	7E-06	6E-06
house	13	0	2.88E-09	0	4E-08	3E-06	4E-07	4E-05	4E-05
laughter	14	0	9.86E-05	0	6E-09	4E-05	3E-05	3E-10	1E-10
light	15	0	0.00806	1	0.102	0.192	0.117	0.8	0.819
music	16	0	3.74E-06	0	3E-10	4E-07	6E-07	8E-12	8E-12
night	17	0	2.59E-07	0	1E-08	6E-06	2E-06	1E-07	7E-08
play	18	0	0.28133	0	6E-04	0.016	0.044	1E-05	2E-05

quiet	19	0	1.75E-08	0	1E-06	5E-05	6E-06	0.002	0.003
smile	20	0	0.028294	0	0.001	0.217	0.098	0.003	0.002
the	21	0	0.091193	0	5E-04	0.039	0.057	6E-05	9E-05
to	22	0	1.27E-09	0	4E-12	1E-10	2E-10	9E-14	4E-14
water	23	0	0.000172	0	1E-06	8E-05	1E-04	2E-07	3E-07

Last Token Position 5

Next Token Probabilities

	t	0	1	2	3	4	5	6	7
[S]	0						0.021		
a	1						9E-06		
and	2						1E-04		
apples	3						0.026		
brings	4						0.007		
by	5						0.004		
cat	6						3E-06		
day	7						2E-07		
dog	8						4E-07		
eat	9						0.002		
fill	10						0.622		
friend	11						4E-07		
happy	12						0.001		
house	13						4E-07		
laughter	14						3E-05		
light	15						0.117		
music	16						6E-07		
night	17						2E-06		
play	18						0.044		
quiet	19						6E-06		
smile	20						0.098		
the	21						0.057		
to	22						2E-10		
water	23						1E-04		